

Garment Workers Productivity Prediction

Final Project

Group 04

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Abstract

This study focuses on predicting the productivity of workers in the garment industry. The garment industry depends heavily on workers, so productivity is very important to meet production targets and delivery deadlines. The dataset used in this study contains information about workers, production conditions and productivity levels.

Different statistical tests and machine learning models were used to understand the factors that affect productivity. The analysis was mainly done for two departments Sewing and Finishing. Techniques such as cluster analysis, correlation analysis, and predictive modeling were used.

The result show that the worker productivity depends on several factors such as incentives, number of workers, SMV (Standard Minute Value), overtime and workload. Different models performed better for different departments. The findings can help garment factories improve productivity and manage workers more effectively.

Table of content

• Abstract	02
• Introduction	03
• Description of the question you are going to answer	03
• Description of the data set	03
• Data Preprocessing	04
• Clustering	04
• Explanatory Data Analysis	06
• Model Selection	12
• Appendix	15

Tables and Figures

- **Table 1 - Variable Summary**
- **Figure 1 - FAMD components**
- **Figure 2 - Silhouette Scores**
- **Figure 3 - Clusters**

Introduction

The garment industry is one of the most important industries in the world. It provides employment to millions of workers and plays a major role in many economics. Since most garment production tasks are done annually, worker productivity is very important.

Factories must maintain high productivity to meet customers demand and deliver products on time. However, many factories can affect worker productivity such as workload, team size, incentives, overtime and production targets.

This study aims to analyze worker productivity using real production data from garment factories. By studying the patterns in the data, we can understand which factors affect productivity the most. Machine learning models are also used to predict productivity levels in different departments.

The main focus of this study is on two departments: **Sewing** and **Finishing**, which are key stages in garment production.

Description of the question

Question: What are the features that affected to the garment worker productivity in Bangladesh?

In this study four objectives were considered to predict worker productivity in the Sewing department ,to predict the worker productivity in the Finishing department, to identify the most important factors affecting productivity and to compare productivity patterns between the two departments.

Description of the dataset

No	Variable Name	Variable Type	Description
1	Date	Date	Date of the observation in MM-DD-YYYY format.
2	Day	Categorical	Day of the week for the production record.
3	Quarter	Categorical	A portion of the month. Each month is divided into four quarters.
4	Department	Categorical	Department associated with the production instance (Sewing or Finishing).
5	Team No	Categorical Identifier /	Identification number assigned to each production team.
6	No_of_Workers	Numerical (Discrete)	Total number of workers in each team.
7	No_of_Style_Change	Numerical (Discrete)	Number of changes in the product style during production.
8	Targeted_Productivity	Numerical (Continuous)	Productivity target set by management for each team per day.
9	SMV	Numerical (Continuous)	Standard Minute Value – the standard time allocated to complete a task.

10	WIP	Numerical (Discrete)	Work in Progress – number of unfinished items in the production line.
11	Over_Time	Numerical (Continuous)	Total overtime worked by the team, measured in minutes.
12	Incentive	Numerical (Continuous)	Financial incentive provided to workers to motivate performance (in BDT).
13	Idle_Time	Numerical (Continuous)	Time when production was interrupted due to various reasons.
14	Idle_Men	Numerical (Discrete)	Number of workers who were idle during production interruptions.
15	Actual_Productivity	Numerical (Continuous)	Actual productivity achieved by workers, ranging from 0 to 1.

Table 1: Variable Summary

Data Preprocessing

First the dataset was checked for missing values and incorrect data types. The date variable was converted into a proper data format, and variables such as day, quarter, and department were treated as categorical variables.

Next, unnecessary or less important variables such as team no was removed because they had very little impact on productivity according to the analysis.

The data set included 506 missing values in the variable WIP (work in progress) so the dataset was splitted in to two and the mice imputation was used for the training data and the same algorithm was used for the testing data.

Clustering

Since the dataset contains both numeric and categorical features, **FAMD was applied to training data without the actual productivity to transform the mixed data into numeric components, and K-Means clustering was performed on these components.**

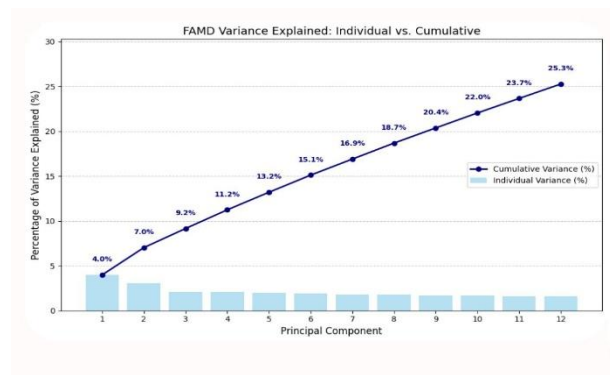


Figure 2: FAMD components

The cumulative variance didn't reach 100% or it hadn't even reached 84% or above. So, all the FAMD components were used in order to take the clusters otherwise there can be information lost. After that the silhouette scores were taken k from 2 to 7 and the below results were obtained. And all the values are less than 0.5. In order to visualize it, 2D plot was plotted against the first 2 FAMD components for k=2.

Therefore, for the modelling part two clusters were considered. From the cluster-wise summary it was clear that those clusters represent the departments Sewing and Finishing.

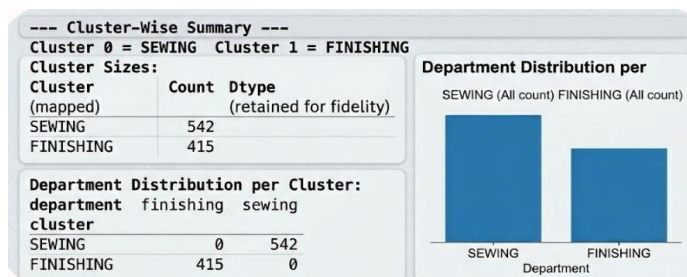
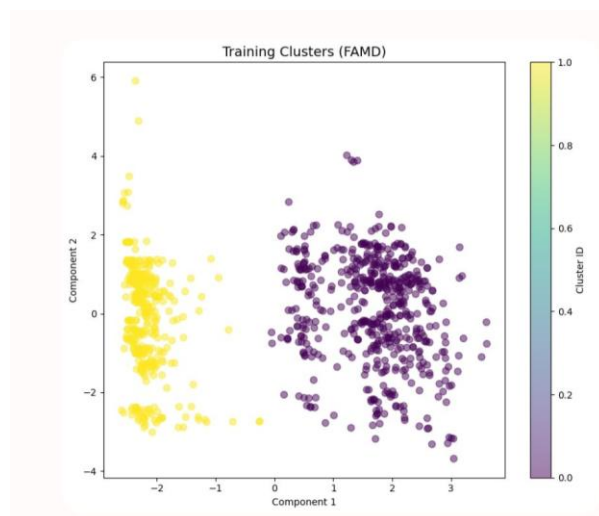
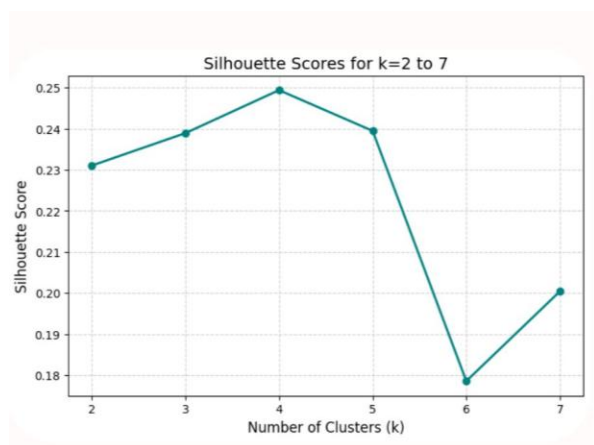
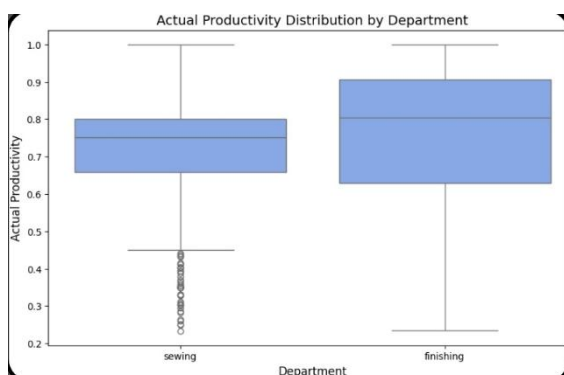


Figure 4: Clusters

Figure 3: Silhouette Scores



Actual Productivity distribution among two clusters



The boxplot compares actual productivity between the sewing and finishing departments. The finishing department shows a slightly higher median productivity, while the sewing department contains several lower outliers. This indicates that productivity in the finishing department is generally higher and more variable than in the sewing department.

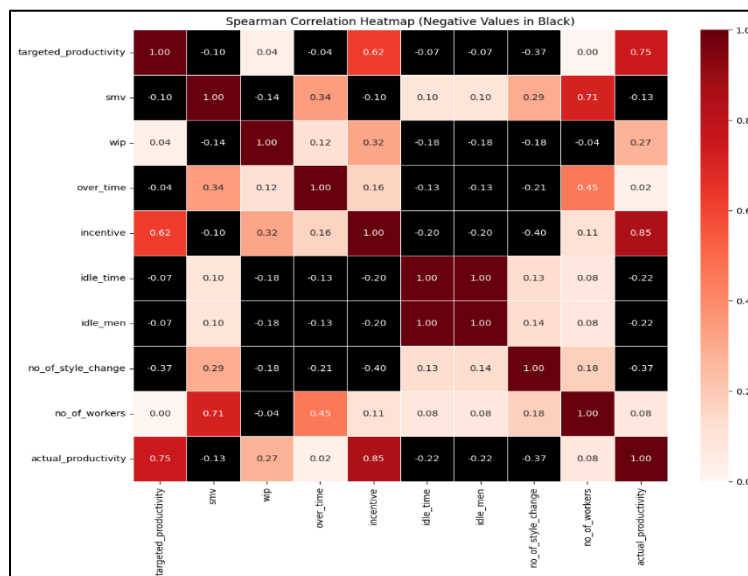
Explanatory Data Analysis.

Exploratory Data Analysis (EDA) is carried out separately for the sewing and finishing departments because they represent two distinct groups with different work characteristics. Analyzing them separately helps in understanding the patterns, variability, and behavior of factors like overtime, incentives, and productivity within each department. The EDA results revealed differences in distribution and workload patterns between the two departments.

To statistically support these observations, we used non-parametric tests. We applied the Shapiro-Wilk test to check whether the data was normally distributed (**H₀: data is normally distributed, H₁: data is not normally distributed**). Since the data was not normally distributed, we used the Mann-Whitney U test to compare the distributions between the two departments (**H₀: both departments have the same distribution, H₁: the distributions are different**). We also used Spearman's correlation to measure the relationship between variables, as it works well with non-normal data. This test allows us to capture both linear and monotonic non-linear relationships between variables (**H₀: There is no relationship between variables, H₁: There is a relationship between variables**). These statistical tests help validate the patterns identified during the EDA.

Explanatory Data Analysis for Sewing Department

Spearman Correlation Heatmap

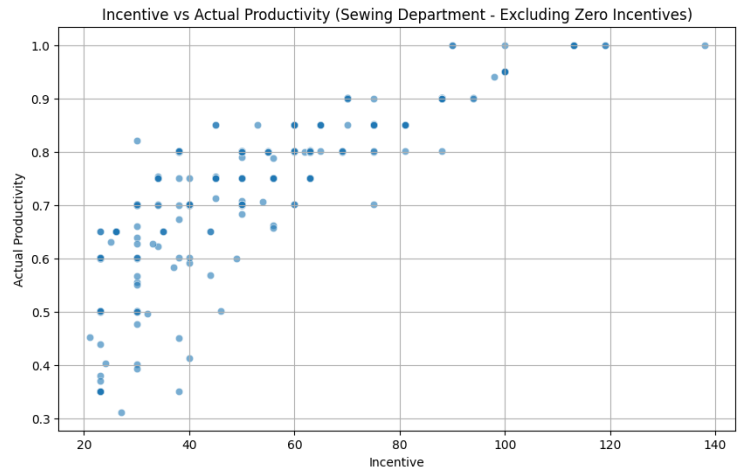


The Spearman correlation heatmap shows that actual productivity has a strong positive relationship with incentive (0.85) and targeted productivity (0.75). So, higher incentives and targets lead to higher productivity. WIP (0.27) has a weak positive relationship with actual productivity. Number of style changes (-0.37) and idle time/idle men (-0.22) have negative relationships with productivity, meaning they reduce performance.

SMV is strongly correlated with the number of workers (0.71) and moderately with overtime (0.34). Overall, incentives and targeted productivity improve the actual productivity, while operational interruptions such as style changes and idle time/men decrease it.

Incentive Vs actual Productivity

This scatter plot provides a visual confirmation of the 0.85 Spearman correlation we saw earlier. After excluding the 0 incentive to identify this pattern clearly, we can see a **positive non-linear relationship**, meaning that as the **incentive amount increases, the actual productivity of workers also tends to increase**. Most of the points move from the lower-left region to the upper-right region of the graph, which visually suggests a positive association between the two variables.



Sewing - Spearman Correlation: 0.8512, p-value: 2.7565e-153

The **p-value is 2.7565×10^{-153}** , which is **much smaller than the significance level of 0.05**. Therefore, we **reject the null hypothesis (H_0)**. So, we can ensure that there is a relationship between the variables. Overall, this result suggests that the **incentive is highly effective in motivating workers and improving their productivity** in the sewing department. Providing higher incentives appears to encourage employees to increase their output, which positively contributes to production performance.

No of workers Vs SMV

The scatter plot illustrates the relationship between the number of workers and Standard Minute Value (SMV) in the sewing department. To evaluate the strength and direction of this relationship, a Spearman correlation analysis was conducted.



Sewing - Spearman Correlation: 0.7110, p-value: 1.3165e-84

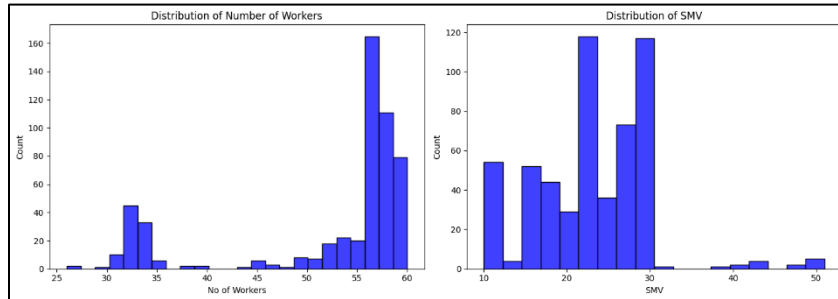
The results show a Spearman correlation coefficient of 0.711, with a p-value of 1.3165×10^{-84} , which is significantly lower than the significance level of 0.05. This means there is a relationship between the two variables.

The correlation coefficient of 0.711 suggests a fairly strong positive relationship between the number of workers and SMV. In simple terms, as the SMV increases, the number of

workers assigned to the sewing line also tends to increase. Since SMV represents the time needed to complete a garment operation, higher SMV values usually correspond to more complex or time-consuming tasks.

Therefore, these results suggest that more complex sewing operations need a larger workforce to maintain production efficiency and meet output targets. This matches production management practices where labor allocation changes based on the complexity and workload of the sewing operations.

Investigating the number of workers and Standard Minute Value (SMV)



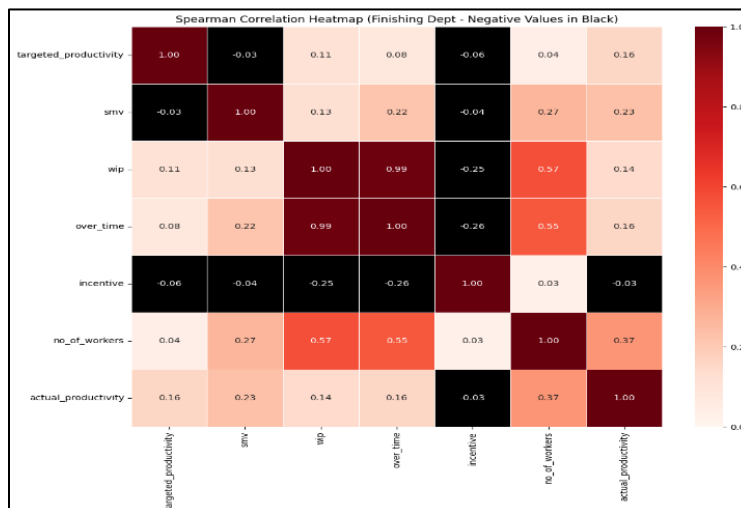
The above histograms illustrate the distribution of the number of workers and the Standard Minute Value (SMV) in the sewing department. Range of the No. of workers is 26-60 while Range of the SMV is 10 to 51 minutes.

The distribution of the number of workers shows that most production lines operate with approximately 55–60 workers. This indicates that the majority of sewing lines are staffed with a relatively similar workforce size, suggesting a standardized production line structure within the factory. A smaller number of observations fall in the **30–40 worker range**, which may refer to smaller production lines or less complex work operations.

The SMV distribution shows that most garment operations fall within the 20–30 SMV range, indicating that the majority of sewing tasks require moderate production time and complexity. There are fewer operations with very low SMV values (around 10–15) and very high SMV values (above 40), suggesting that extremely simple or highly complex tasks are less common.

This pattern suggests that the factory generally operates with similar line sizes and moderate task complexity, which may help maintain consistent production efficiency and easier line balancing.

Explanatory Data Analysis for Finishing Department. Spearman Correlation Heatmap



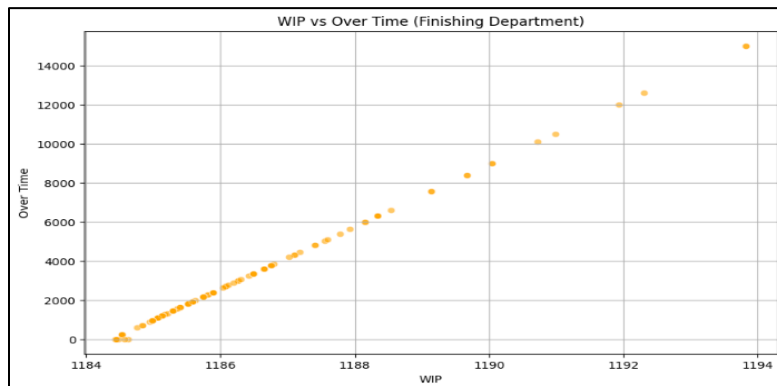
The Spearman correlation heatmap shows the relationships among key production variables in the finishing department. The strongest relationship is between WIP and overtime ($r = 0.99$), indicating that higher work-in-progress leads to increased overtime.

There is also a moderate positive relationship between WIP and number of workers ($r = 0.57$) and

overtime and number of workers ($r = 0.55$), suggesting that higher workload requires more workers and longer working hours.

Additionally, the number of workers has a moderate positive correlation with actual productivity ($r = 0.37$), indicating that increasing the workforce is important to get higher productivity. Most other variables show weak relationships, meaning their influence on productivity is relatively low.

WIP Vs Over time



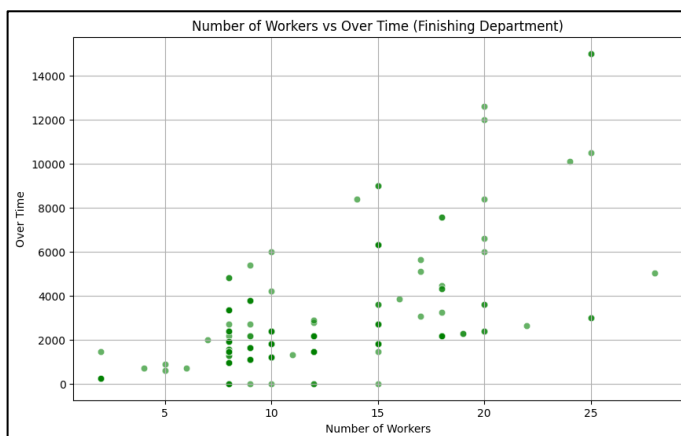
Spearman Correlation (WIP vs Over Time - Finishing):
Correlation coefficient: 0.9861
p-value: 0.0000e+00

The scatter plot shows the relationship between **Work in Progress (WIP)** and **Overtime hours** in the finishing department. From the graph, it is clear that as **WIP increases, overtime hours also increase**. The points follow a strong upward trend, indicating a strong positive linear relationship between the two variables. The **Spearman correlation coefficient is 0.9861**, which is very close to 1, showing a **very strong positive**

correlation between WIP and overtime. In addition, the **p-value is less than 0.05**, which means that there is a relationship.

This result suggests that when the amount of unfinished work waiting in the finishing department increases, employees need to **work more overtime hours to complete the workload**. In other words, higher WIP leads to increased overtime requirements in order to maintain production flow and meet deadlines.

Overtime vs no of workers



Spearman Correlation (Finishing Department):
Correlation coefficient: 0.5500
p-value: 3.4892e-34

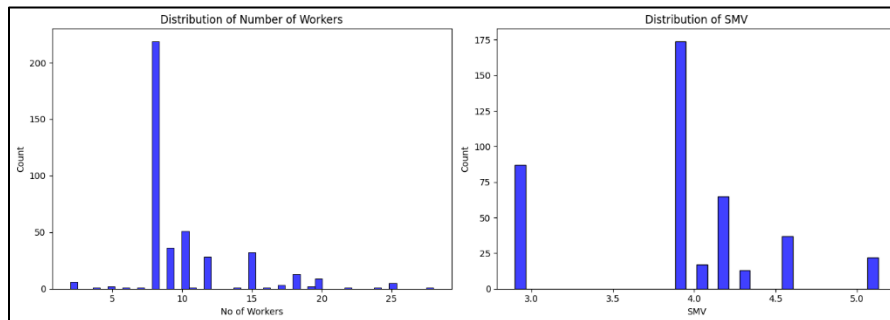
The scatter plot shows the relationship between the number of workers and overtime hours in the finishing department. The Spearman correlation coefficient is 0.5500, indicating a moderate positive relationship between these two variables. The p-value (3.4892e-34) is less than 0.05, which means that There is a relationship

between these variables. However, the points on the scatter plot are widely spread, showing that

the relationship is not very strong. In several cases, even when the number of workers is high, the overtime hours are still high. This suggests that **increasing the number of workers alone does not necessarily reduce overtime**.

This may happen because overtime is influenced by other factors such as **high workload, production targets, process inefficiencies, or delays in earlier production stages**. Although there is a **significant moderate non-linear relationship between the number of workers and overtime**, simply adding more workers does **not guarantee a reduction in overtime**, especially when the workload remains high.

Investigating the number of workers and Standard Minute Value (SMV)



The histograms display the number of workers and Standard Minute Value (SMV) in the finishing department. The number of workers mostly falls between 8 and 12, showing that this is the usual workforce

size in the finishing department. The total range goes from 2 to 28 workers. However, some teams have 25 to 27 workers, which likely happens during times of heavy workload or when extra help is needed to meet production goals.

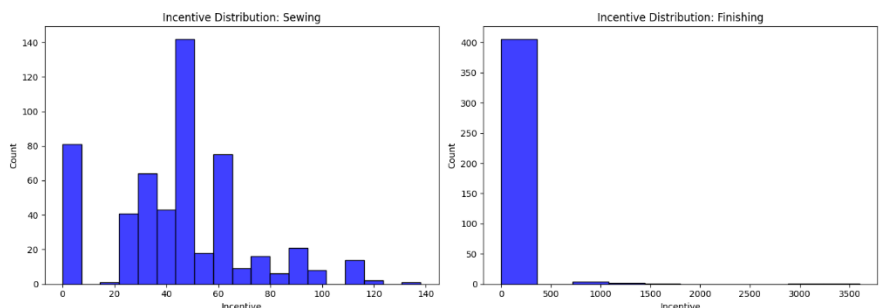
The SMV distribution reveals that the majority of operations take about 3.8 to 4.2 minutes. This suggests that most tasks require about the same amount of time. A few higher SMV values, around 4.5 to 5.1, point to more complex or time-consuming tasks.

Overall, the distributions show that while the workforce size changes based on the workload, the SMV values are quite similar. This means that most finishing operations need about the same amount of processing time.

Exploratory Data Analysis for Comparison of Two Departments

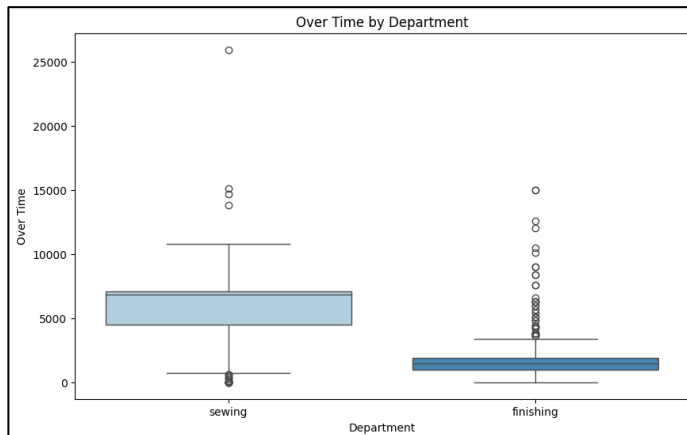
Incentive

The histogram shows differences in incentive distribution between the sewing and finishing departments. Sewing incentives are mostly concentrated between 30 and 70, indicating a relatively consistent distribution among worker. In contrast, the finishing department shows a highly



skewed distribution with many small values and a few extremely large incentives. These large values create outliers and suggest irregular incentive allocation. Overall, sewing incentives are more stable, while finishing incentives show greater variability.

Over Time



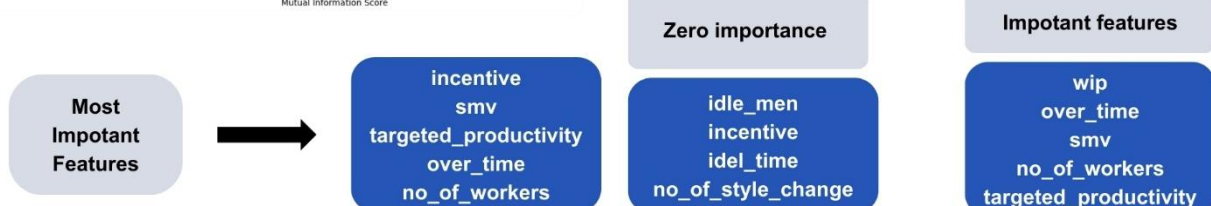
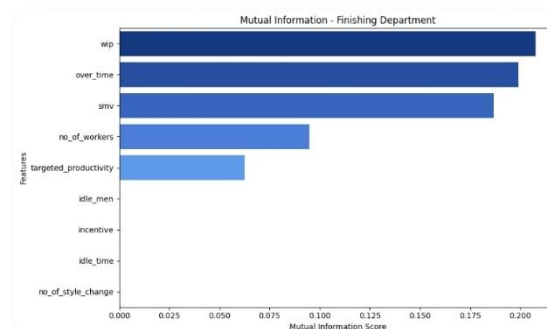
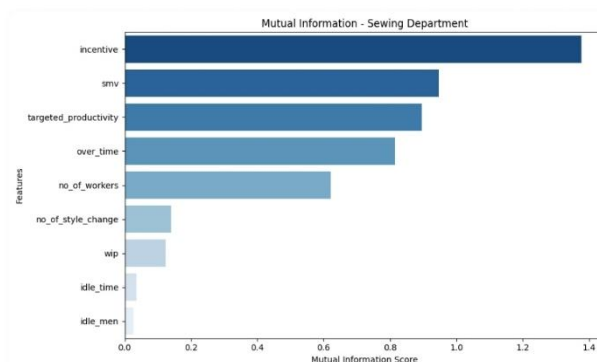
The boxplot shows that the sewing department generally has higher overtime compared to the finishing department. The sewing department also has a wider spread, indicating greater variability in overtime hours. But, finishing overtime is mostly concentrated at lower values but includes several outliers. These outliers suggest that occasionally some finishing workers work unusually high overtime. Overall, sewing has higher and more variable overtime than finishing.

Shapiro-Wilk (Sewing) - Over Time: Statistics=0.9266, p-value=1.2646e-15
 Shapiro-Wilk (Finishing) - Over Time: Statistics=0.5811, p-value=2.2178e-30
 Mann-Whitney U Test - Over Time: Statistics=204335.0000, p-value=1.6808e-104

The data show that overtime hours in both the sewing and finishing departments

don't follow a normal distribution, as indicated by the Shapiro-Wilk test ($p < 0.05$). The Mann-Whitney test results ($p < 0.05$) suggest a statistically significant difference in overtime between the two departments. In general, the sewing department has more overtime hours, and these hours also vary more than in the finishing department.

Mutual Information for the both departments



None of the variables was dropped as these variables are essential in a domain of garments although the data was quite less in the period of data collecting. Tree based models were considered later because it handles everything well so not removing these variables won't be much problem.

Selection Of the Models

Multicollinearity Considerations

The Correlation analysis revealed that several predictors, particularly **Target productivity, SMV, and no of workers**, were moderately to highly correlated in the finishing department. This indicates the presence of multicollinearity, which can inflate variance of coefficient estimates in linear models. Since removing variables would result in loss of valuable information. We have to use either regularized linear models or tree-based methods that can handle correlated predictors effectively. We use below models to select best model. Ridge Regression, Lasso Regression, Elastic Net Regression, XGBoost Model, Random Forest Model.

Model Selection Strategy

To identify the most suitable predictive model, both training and testing performances were evaluated using several error metrics, namely Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2). MSE and RMSE measure the average squared prediction error, giving more weight to larger deviations, while MAE provides the average magnitude of errors in a more interpretable scale. R-squared indicates the proportion of variability in response variability explained by the model.

By comparing these metrics across training and testing datasets, we assessed not only the accuracy but also the generalization ability of each model. A good model is expected to have low error values and a high R-squared in test data, with minimal difference between training and testing results.

Fit the models for the Sewing Department and Finishing Department

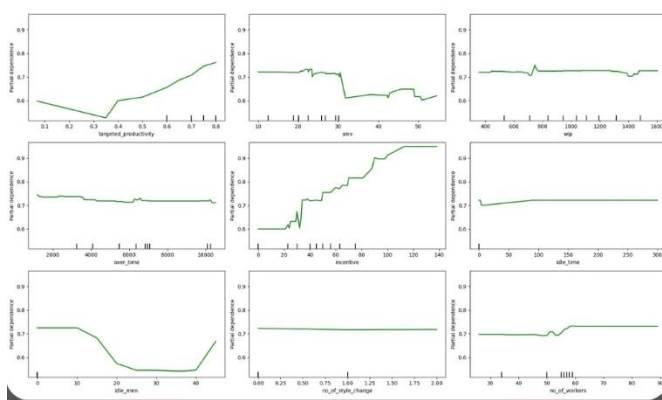
<i>Table 4: Model summary (Sewing)</i>					<i>Table 5: Model summary (Finishing)</i>				
Model	Train R^2	Test R^2	Train MSE	Test MSE	Model	Train R^2	Test R^2	Train MSE	Test MSE
XGBoost	0.9488	0.8015	0.0013	0.0041	Random Forest	0.4383	0.2310	0.0217	0.0269
Random Forest	0.8920	0.7598	0.0027	0.0049	XGBoost	0.3796	0.1772	0.0240	0.0288
Lasso	0.8062	0.7727	0.0048	0.0047	Lasso	0.1470	0.0994	0.0330	0.0315
Ridge	0.8083	0.7686	0.0047	0.0048	Ridge	0.1425	0.1169	0.0331	0.0309
ElasticNet	0.8036	0.7676	0.0048	0.0048	ElasticNet	0.1322	0.1293	0.0335	0.0305
Several models were tested, including XGBoost, Random Forest, Lasso, Ridge, and					For the Finishing department, the same models were tested. The Random Forest model				

Elastic Net. Among these models, XGBoost performed the best. It achieved the highest Test R^2 value of 0.8015, which means it explains about 80% of the variation in productivity. It also produced the lowest testing error (Test MSE = 0.0041) compared to the other models. Random Forest also performed well but had slightly lower accuracy than XGBoost. Linear models such as Lasso, Ridge, and ElasticNet showed lower predictive performance. **Therefore, XGBoost was selected as the best model for predicting productivity in the Sewing department.**

performed the best with a Test R^2 value of 0.2310 and a Test MSE of 0.0269. This indicates that Random Forest predicts finishing productivity better than the other models. Other models such as XGBoost, Lasso, Ridge, and ElasticNet showed lower prediction performance. **Therefore, Random Forest was selected as the best model for the Finishing department.**

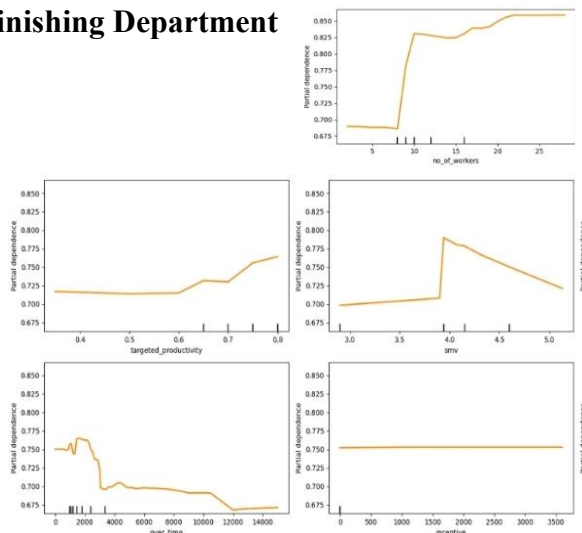
Feature Impact Analysis Using Partial Dependence Plots

Sewing Department



The partial dependency plots illustrate the impact of various elements on productivity in the sewing department. Factors like set productivity goals and incentives positively influence productivity, indicating that increased targets and improved incentives motivate employees to enhance their performance. The number of employees also somewhat boosts productivity, but beyond a certain threshold, the impact levels off. Conversely, excessive SMV and overtime can hinder productivity as tasks require more time and workers may experience fatigue.

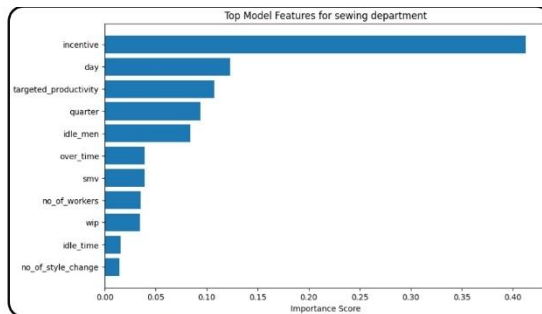
Finishing Department



In the finishing department, productivity is mainly influenced by targeted productivity, number of workers, overtime, and WIP (work in progress). Increasing the number of workers and having clear productivity targets generally improves productivity. However, higher SMV and excessive overtime may lower productivity because tasks become more time-consuming and workloads increase. Overall, finishing productivity depends mostly on workload and team size.

Feature Importance Analysis of the Final Model

XG boost – Sewing Department



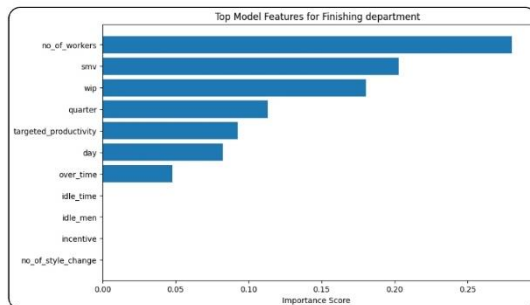
Important features

incentive
smv
targeted_productivity
over_time
no_of_workers

These features were obtained from
the mutual information too.

For the **Sewing Department**, the **XGBoost model** identified several features that strongly influence productivity. Among them, **incentive**, **SMV**, **targeted productivity**, **overtime**, and **number of workers** appear as the most important variables. The feature importance plot shows that incentives have the highest contribution to the model's predictions, followed by production-related variables such as targeted productivity and SMV. These results indicate that both workforce management and production targets play a key role in determining sewing department performance. Notably, similar important features were also identified using the **mutual information** method, which supports the reliability of the selected variables.

Random Forest – Finishing Department



Important features

wip
over_time
smv
no_of_workers
targeted_productivity

Similar features were obtained from
the mutual information too.

For the **Finishing Department**, the **Random Forest model** highlights a slightly different set of influential variables. The most important features include **WIP (work in progress)**, **overtime**, **SMV**, **number of workers**, and **targeted productivity**. The feature importance plot shows that operational workload and workforce size significantly impact finishing department productivity. The consistency between the Random Forest feature importance results and the **mutual information analysis** suggests that these variables are strong predictors of productivity and are relevant for building reliable machine learning models.

Conclusion

This study analyzed worker productivity in the garment industry using statistical analysis and machine learning techniques. The results show that several factors influence productivity, including targeted productivity, SMV, number of workers, incentives, overtime, and work in progress (WIP). The analysis also revealed that the sewing and finishing departments have different productivity patterns, mainly because their tasks and workflows are different.

Different predictive models were tested to find the most suitable model for each department. The XGBoost model performed best for the sewing department, while the Random Forest model performed best for the finishing department. These results show that using appropriate models for different departments can improve prediction accuracy. The findings of this study can help garment factory managers make better decisions about workforce management, workload planning, and incentive strategies to improve overall productivity.

References

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- <https://www.mdpi.com/1424-8220/23/21/8694>
- <https://medium.com/@mbkioko/employee-performance-analysis-d5bf904b8039>

Appendix

Dataset

[Productivity Prediction of Garment Employees](#)

Google Collab link

https://colab.research.google.com/drive/1_c8csrvDfPSl4lOxPi6VvBSzuU_MvdxJ?usp=sharing

Python File, Website Video Demonstration, Dataset file

https://drive.google.com/drive/folders/1ZJ_tAdrYwVPq1BzvaX5WV5CN63daLLG4?usp=sharing